Algebra II Polynomial Functions Cheatsheet

Module 4

End Behavior

Use degree and leading coefficient.

even degree	positive leading coefficient	both ends up
even degree	negative leading coefficient	both ends down
odd degree	positive leading coefficient	left down, right up
odd degree	negative leading coefficient	left up, right down

Zeros and Multiplicity

If

$$f(x) = a(x - r)^m(\cdots)$$

then r is a zero with multiplicity m.

$$m \text{ even} \Rightarrow \text{touches and turns}$$

$$m \text{ odd} \Rightarrow \text{crosses}$$

Rational Root Theorem

If

$$f(x) = a_n x^n + \cdots + a_0$$

has rational root $\frac{p}{q}$ in lowest terms, then

$$p \mid a_0, \qquad q \mid a_n$$

Possible rational roots:

$$\pm\frac{\text{factors of constant term}}{\text{factors of leading coefficient}}$$

Remainder and Factor Theorems

Remainder Theorem:

$$\text{remainder on division by } x - r \text{ is } f(r)$$

Factor Theorem:

$$f(r) = 0 \iff x - r \text{ is a factor}$$

Division

Use synthetic division only for divisors of the form

$$x - r$$

Use long division for other polynomial divisors.

Complex Roots

For real-coefficient polynomials:

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Nonreal roots come in conjugate pairs.

Graph Checklist

- Find real zeros and multiplicities.
- Find y-intercept $f(0)$.
- Determine end behavior.
- Use turning points and sign checks to refine shape.